Formal Verification of Multivariate Polynomial Factorization over $\mathbb{Q}$

The Computability Barrier of the Reals

Mechanizing real-valued algorithms introduces a fundamental clash between classical mathematics and computational logic. In formal proof systems, the continuous real numbers ($\mathbb{R}$) are typically constructed non-constructively (e.g., via Cauchy sequences or Dedekind cuts). Because true real numbers have infinite precision, exact arithmetic over arbitrary reals is not directly computable, and equality testing ($a = 0$) is undecidable.

To perform mathematically verified factorization, the problem must be shifted to a computable domain such as $\mathbb{Q}[x, y, z]$. Because $\mathbb{Q}$ is a field, it guarantees exact computability. By iterative application of Gauss's Lemma, $\mathbb{Q}[x, y, z]$ is mathematically guaranteed to be a Unique Factorization Domain (UFD), shifting the problem back into the realm of classical, executable algebra.

Algorithmic Approaches over $\mathbb{Q}$

1. Kronecker Substitution

This approach uses an injective ring homomorphism to pack a multivariate polynomial into a massive univariate polynomial, which can then be factored using existing verified univariate libraries.

- **Determine Degree Bounds:** Calculate a degree bound d that strictly exceeds the degree of $f(x, y, z)$ in any single variable.
- **Apply Substitution Mapping:** Map $f(x, y, z)$ to $F(w) \in \mathbb{Q}[w]$ using the substitution $x \mapsto w$, $y \mapsto w^d$, $z \mapsto w^{d^2}$. This establishes a strict bijection between the multivariate terms and the resulting univariate terms, ensuring the mapping preserves factorability.
- **Univariate Factorization:** Execute a verified univariate factorization algorithm on $F(w)$ over $\mathbb{Q}[w]$.
- **Inverse Mapping and Recombination:** Apply the inverse mapping via base-d expansion of the univariate exponents to pull the factors back to $\mathbb{Q}[x, y, z]$. Because the substitution can artificially split irreducible factors, recombine and test candidate factors via polynomial division.

2. Multivariate Hensel Lifting

Instead of inflating the degree, this method projects the polynomial down to a univariate domain, factors it, and iteratively lifts the factors back up.

- **Variable Evaluation:** Choose an evaluation ideal $I = \langle y - a, z - b \rangle$ for integers a, b such that the projected polynomial $f(x, a, b) \in \mathbb{Q}[x]$ remains square-free and retains the exact same total degree in x.
- **Base Factorization:** Factor $f(x, a, b)$ univariately over $\mathbb{Q}$.
- **Hensel Lifting:** Iteratively lift the factorization modulo I^k for increasing k until k exceeds the total degree bound of the original multivariate polynomial.
- **Recombination:** Group the lifted tentative factors to find the true irreducible components in $\mathbb{Q}[x, y, z]$.